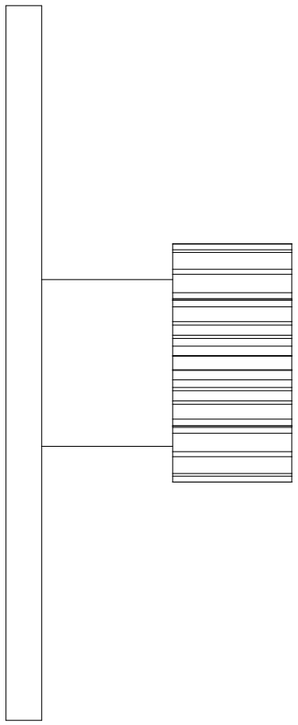
VAYN shoulder · milling drawing · 22 Carrier-1 front and sun-2

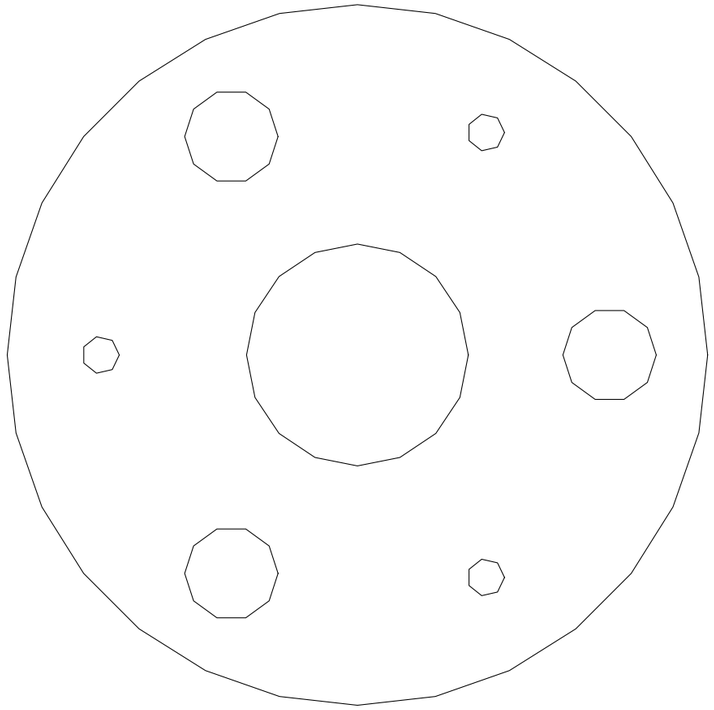
Pack A · geometry Rev 1.4 · qty 1 actuator

FRONT · Z horizontal, +X up



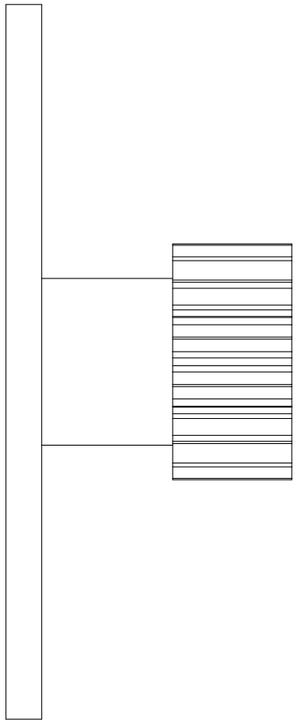
scale 1.94:1 view span 24.0 × 60.0 mm

END · from the arm side (+Z)



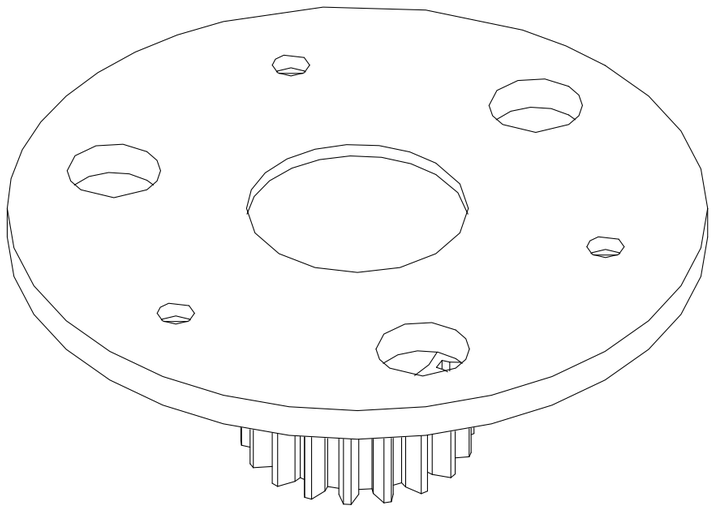
scale 1.90:1 view span 60.0 × 60.0 mm

SIDE · Z horizontal, +Y up



scale 1.94:1 view span 24.0 × 60.0 mm

ISOMETRIC



scale 1.90:1 view span 60.0 × 42.6 mm

22

Carrier-1 front and sun-2

Quantity in this actuator: 1
Material: 16MnCr5
Group: Gears
Finished size: Plate Ø60 mm, sun Ø20 × 10 mm face, overall about 24 mm
Mass: 92.03 g each
Model bounding box, X × Y × Z: 60.0 × 60.0 × 24.0 mm
Raw stock: Ø65 × 35 mm 16MnCr5, annealed, same heat
Process: Turn, hob, drill
Heat treatment / finish: Carburize the sun only. Case 0.15–0.20 mm, 58–62 HRC. Mask the plate.
STEP solid: 22 Carrier-1 front and sun-2 (16MnCr5)

Controlled features

- Sun-2: 18 teeth, module 1.0. Span over 3 teeth W3 = 7.609 mm, minus 0.015 / plus 0.
- ISO 1328 grade 6–7.
- 3 × Ø8 mm R7 holes for the planet pins.
- 3 × M3 countersunk on Ø44 mm.
- Rear recess Ø19 × 1.2 mm, so the sun-1 tips clear by 1.5 mm.

STEP file is the geometric master. Do not scale this sheet. Sizes not listed here follow ISO 2768-1 class m and ISO 2768-2 class K. Drawing 22_carrier_1_front.pdf